SIAMIS ICT Weekly Lesson Plan

Teacher: Ray Joshua Balingkit | Year 10–11 | Term 2 | Week 3

Year 10 course sequence: Units 1–3. Two 45-minute lessons. Year 11 progression: Units 4–6.

Lesson overview

This week covers transmission media, hubs and switches and routers, boosters and servers.

Pearson Edexcel International GCSE (9-1) ICT Student Book, printed pages 83-86

End of week assessment

Check accurate terminology, explanations and application through the relevant book activities and questions. Record misconceptions and use the next feedback or consolidation session for corrections. Assess only content already taught.

Lesson 1

Transmission media, hubs and switches

Date: 18 January 2027 | Duration: 45 minutes

Learning objective

By the end of this lesson, students will be able to explain and apply transmission media, hubs and switches, using accurate textbook terminology and a justified example.

Lesson flow

Hook 5 minutes

Use a short scenario, image or retrieval question to introduce transmission media, hubs and switches.

Introduce 8 minutes

Teach transmission media, hubs and switches from printed pages 83-86; include the relevant definitions, diagrams and examples.

We do 10 minutes

Work through a textbook example or comparison for transmission media, hubs and switches; explain each decision and check understanding.

You do 17 minutes

Complete the relevant textbook activities and questions for transmission media, hubs and switches. Use a practical demonstration or scenario where appropriate; finish remaining practice as homework.

Check 5 minutes

Explain one key idea about transmission media, hubs and switches and justify an answer; record misconceptions for the next lesson.

Differentiation

Support: Provide labelled diagrams, vocabulary, a structured response frame and a worked example. Apply agreed access arrangements during independent assessment without supplying answers.

Extension: Justify an alternative solution, explain a limitation or apply the same concept to a new scenario using evidence.

Lesson 2

Routers, boosters and servers

Date: 20 January 2027 | Duration: 45 minutes

Learning objective

By the end of this lesson, students will be able to explain and apply routers, boosters and servers, using accurate textbook terminology and a justified example.

Lesson flow

Hook 5 minutes

Use a short scenario, image or retrieval question to introduce routers, boosters and servers.

Introduce 8 minutes

Teach routers, boosters and servers from printed pages 83-86; include the relevant definitions, diagrams and examples.

We do 10 minutes

Work through a textbook example or comparison for routers, boosters and servers; explain each decision and check understanding.

You do 17 minutes

Complete the relevant textbook activities and questions for routers, boosters and servers. Use a practical demonstration or scenario where appropriate; finish remaining practice as homework.

Check 5 minutes

Explain one key idea about routers, boosters and servers and justify an answer; record misconceptions for the next lesson.

Differentiation

Support: Provide labelled diagrams, vocabulary, a structured response frame and a worked example. Apply agreed access arrangements during independent assessment without supplying answers.

Extension: Justify an alternative solution, explain a limitation or apply the same concept to a new scenario using evidence.

Materials

Teacher computer and display

School-approved applications, class save folder and fictional scenarios

Relevant textbook diagrams, activities and chapter questions

Student notebooks and a teacher assessment checklist

Teacher reflection

What worked well? _____

What needs reteaching or adjustment? _____